

Customer Segmentation Assistant

(AI-ML Based Customer Segmentation Platform)

A PROJECT REPORT

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ABSTRACT

This project, Customer Segmentation Assistant, is an AI-ML web application developed to identify meaningful groups of mall customers using unsupervised machine learning. The system uses the Mall Customers dataset containing 200 records with demographic and spending-related attributes. After data cleaning, exploratory analysis, feature scaling, and model selection, the K-Means algorithm is applied to segment customers based on annual income and spending score.

The final model uses $K = 5$ clusters and achieves a silhouette score of 0.5547, indicating clear separation between customer groups. The identified segments include budget shoppers, impulsive buyers, careful spenders, average customers, and target customers. The application also includes a Flask dashboard, visualizations, cluster interpretation, marketing recommendations, and new-customer prediction support.

INTRODUCTION

Customer segmentation is an important business analytics technique used to divide customers into groups with similar purchasing behaviour. Retail organizations can use these groups to personalize offers, improve customer retention, and make better marketing decisions.

The Customer Segmentation Assistant applies machine learning to transform raw customer records into actionable business insights. Instead of manually assigning customer categories, the system uses K-Means clustering to discover natural patterns in income and spending behaviour.

The project is implemented as a Python Flask web application with pages for dataset overview, EDA, cleaning, scaling, clustering, visualization, interpretation, recommendations, dashboard analytics, prediction, and report generation.

PROBLEM STATEMENT

Businesses often collect customer data but struggle to convert it into practical marketing decisions. Treating all customers in the same way leads to generic promotions, poor targeting, and missed revenue opportunities.

The problem addressed in this project is to automatically group mall customers into meaningful segments using their annual income and spending score. The system must also present the results in a simple dashboard so that the segmentation output can be understood by non-technical users.

OBJECTIVES

- To load and inspect the Mall Customers dataset.
- To perform exploratory data analysis on age, gender, income, and spending score.

- To clean the dataset by removing unnecessary identifiers and checking missing values.
- To scale numeric features before applying K-Means clustering.
- To determine a suitable number of clusters using elbow and silhouette analysis.
- To train a K-Means model and interpret the resulting customer groups.
- To build a Flask dashboard for visualization, prediction, and report generation.
- To provide marketing recommendations for each customer segment.

SCOPE

The current scope of the project includes data preprocessing, EDA, clustering, visualization, segment profiling, marketing recommendation generation, and prediction of the cluster for a new customer based on income and spending score.

The project works on the Mall Customers dataset and demonstrates an end-to-end AI-ML workflow suitable for academic demonstration. It can be extended in the future by adding real retail transaction data, product-level behaviour, visit frequency, CRM integration, and deployment to a public cloud server.

SYSTEM DESIGN & ANALYSIS

The system is designed as a modular Flask application. The user interacts with web pages rendered through HTML templates. The backend loads the dataset, performs preprocessing, trains or loads the machine learning model, and returns visual or tabular outputs to the dashboard.

Layer	Component	Purpose
User Interface	HTML, CSS, Bootstrap	Displays pages and forms
Application Layer	Flask routes	Controls dashboard navigation
Data Layer	Pandas, CSV, SQLite	Loads dataset and stores predictions
ML Layer	StandardScaler, KMeans	Scales features and creates clusters
Visualization	Plotly, Matplotlib	Shows charts and model results

TECHNOLOGIES USED

Technology	Use in Project
Python	Core programming language
Flask	Web application framework
Pandas and NumPy	Data handling and numerical processing

Technology	Use in Project
Scikit-learn	Feature scaling, K-Means clustering, silhouette score
Plotly	Interactive dashboard visualizations
Matplotlib and Seaborn	Report charts and statistical plots
SQLite	Storage of new predicted customers
ReportLab	PDF project report generation
HTML, CSS, JavaScript	Frontend templates and interactions

ARCHITECTURE DIAGRAM

Input / Module	Output / Connected Module
User	Flask Web App
Flask Web App	Preprocessing Module
Preprocessing Module	EDA and Feature Scaling
EDA and Feature Scaling	K-Means Clustering
K-Means Clustering	Cluster Profiles
Cluster Profiles	Dashboard, Prediction, PDF Report

The architecture follows a sequential machine learning pipeline. Data flows from the uploaded or stored CSV file to preprocessing, scaling, model training, cluster interpretation, and finally to user-facing pages such as dashboard, prediction, and report output.

MODULE DESCRIPTION

Module	Description
preprocessing.py	Loads data, removes duplicate rows, renames columns, encodes gender, and scales features.
eda.py	Generates descriptive statistics, missing-value checks, outlier checks, and correlations.
clustering.py	Computes WCSS, silhouette scores, trains K-Means, and predicts new clusters.
visualization.py	Creates charts for age, income, spending, elbow method, silhouette score, and clusters.
recommendations.py	Stores segment profiles and marketing strategies.
database.py	Handles SQLite operations for predicted customer records.
app.py	Defines Flask routes and connects all modules to web pages.

IMPLEMENTATION

The implementation begins with data loading and inspection. CustomerID is dropped because it is only an identifier and does not help clustering. Annual income and spending score are selected as the final clustering features because they represent purchasing power and buying behaviour.

StandardScaler is applied before K-Means because distance-based algorithms are sensitive to feature ranges. The elbow method and silhouette score are used to validate $K = 5$ before fitting the final clustering model.

Metric	Value
Total records	200
Final records after cleaning	200
Missing values	0
Duplicate rows removed	0
Selected features	Annual_Income, Spending_Score
Number of clusters	5
Silhouette score	0.5547
WCSS / Inertia	65.5684

INTERNAL SEGMENTATION WORKFLOW

1. Data Input and Inspection

The segmentation process begins when the application loads the Mall_Customers.csv dataset. The raw file contains CustomerID, Gender, Age, Annual Income, and Spending Score. The application first checks the number of rows and columns, verifies data types, counts missing values, and checks duplicate rows. This step confirms whether the dataset is suitable for machine learning.

2. Cleaning and Feature Preparation

CustomerID is removed because it is only a serial number and does not describe customer behaviour. The long column names are renamed to Annual_Income and Spending_Score so they can be used easily in Python code. Gender is encoded as a numeric value for analysis, but the final K-Means clustering uses Annual_Income and Spending_Score because these two variables directly represent purchasing power and shopping behaviour.

Raw Column	Action Performed	Reason
CustomerID	Dropped	Identifier only; not useful for clustering
Gender	Encoded as Male = 1, Female = 0	Useful for analysis and dashboard summaries
Age	Retained for interpretation	Helps describe customer segment profile
Annual Income (k\$)	Renamed and selected	Represents earning capacity
Spending Score (1-100)	Renamed and selected	Represents purchase behaviour

INTERNAL SEGMENTATION WORKFLOW

3. Feature Scaling

K-Means calculates distance between data points. If one feature has a much larger numeric range than another feature, it can dominate the Euclidean distance calculation. Annual income ranges from 15 to 137, while spending score ranges from 1 to 99. To make both features contribute fairly, StandardScaler converts each feature into a z-score.

The formula used by StandardScaler is: $z = (x - \text{mean}) / \text{standard deviation}$. After this transformation, both income and spending score have approximately mean 0 and standard deviation 1.

Feature	Mean	Std Dev	Minimum	Maximum
Annual_Income	60.56	26.26	15	137
Spending_Score	50.20	25.82	1	99

4. How K-Means Works Internally

- Step 1: Choose K initial centroids using k-means++ initialization.
- Step 2: Calculate Euclidean distance from each customer to each centroid.
- Step 3: Assign each customer to the nearest centroid.
- Step 4: Recalculate each centroid as the mean of all customers assigned to that cluster.
- Step 5: Repeat assignment and centroid update until the centroids stop changing significantly.
- Step 6: Store the final cluster labels and centroids for interpretation and prediction.

INTERNAL SEGMENTATION WORKFLOW

5. Choosing the Number of Clusters

The project evaluates multiple K values before finalizing the model. The elbow method calculates WCSS, which measures compactness inside clusters. As K increases, WCSS decreases, but after a point the improvement becomes small. The silhouette score measures how well separated the clusters are. Higher silhouette values indicate better-defined clusters.

K	WCSS	Silhouette Score
1	400.00	-
2	269.69	0.3213
3	157.70	0.4666
4	108.92	0.4939
5	65.57	0.5547
6	55.06	0.5399
7	44.86	0.5281
8	37.23	0.4552
9	32.39	0.4571
10	29.98	0.4432

K = 5 is selected because it gives meaningful business groups and a strong silhouette score of 0.5547. It separates customers into low-income low-spending, low-income high-spending, medium behaviour, high-income low-spending, and high-income high-spending patterns.

INTERNAL SEGMENTATION WORKFLOW

6. Cluster Profiling

After K-Means assigns cluster labels, the system joins each label back to the cleaned customer table. Then it calculates the size, percentage share, average age, average income, average spending score, and gender distribution for each cluster. These statistics convert mathematical clusters into understandable customer personas.

Cluster	Segment	Meaning
0	Average Customers	Moderate income and moderate spending. These customers represent the stable middle group and can be improved through broad engagement.
1	Target Customers	High income and high spending. These are premium customers who are most valuable for retention, loyalty, and personalized offers.
2	Impulsive Buyers	Low income but high spending. These customers respond well to trends, offers, events, and emotionally engaging campaigns.
3	Careful Spenders	High income but low spending. These customers have purchasing capacity but need trust, value assurance, and stronger incentives.
4	Budget Shoppers	Low income and low spending. These customers are price-sensitive and respond best to discounts, bundles, and loyalty benefits.

7. Prediction for a New Customer

For prediction, the user enters annual income and spending score in the web form. The saved StandardScaler transforms the input using the same scale learned from the training data. The saved K-Means model then checks which centroid is closest to the new point and returns the corresponding cluster ID. Finally, the dashboard displays the matching customer persona and marketing suggestion.

CODE SNIPPETS

Data Cleaning and Feature Scaling

```
df = pd.read_csv("Mall_Customers.csv")

df = df.drop_duplicates()

df = df.drop(columns=["CustomerID"])

df = df.rename(columns={
    "Annual Income (k$)": "Annual_Income",
    "Spending Score (1-100)": "Spending_Score"
})

df["Gender_Encoded"] = df["Gender"].map({"Male": 1, "Female": 0})

X = df[["Annual_Income", "Spending_Score"]].values

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

K-Means Model Training

```
kmeans = KMeans(
    n_clusters=5,
    init="k-means++",
    n_init=10,
    random_state=42,
    max_iter=300
)

labels = kmeans.fit_predict(X_scaled)

score = silhouette_score(X_scaled, labels)

df["Cluster"] = labels
```

CODE SNIPPETS

New Customer Prediction

```
def predict_cluster(annual_income, spending_score):  
  
    X_new = np.array([[annual_income, spending_score]])  
  
    X_scaled = scaler.transform(X_new)  
  
    cluster = int(kmeans.predict(X_scaled)[0])  
  
    return cluster
```

Cluster Summary Logic

```
summary = []  
  
for cid in sorted(df["Cluster"].unique()):  
  
    group = df[df["Cluster"] == cid]  
  
    summary.append({  
  
        "cluster": cid,  
  
        "size": len(group),  
  
        "mean_income": group["Annual_Income"].mean(),  
  
        "mean_spending": group["Spending_Score"].mean(),  
  
        "mean_age": group["Age"].mean()  
  
    })
```

CODE SNIPPETS AND EXPLANATION

Flask Route Structure

The Flask app is the controller of the project. Each route calls the required utility functions, prepares the processed data, and sends it to the matching HTML template. This keeps the machine learning logic separate from the user interface.

```
@app.route("/clustering")

def clustering():

    kmeans = get_kmeans()

    summary = get_cluster_summary()

    sil_data = get_silhouette()

    return render_template(

        "clustering.html",

        kmeans=kmeans,

        summary=summary,

        sil_data=sil_data

    )
```

How the Route Works

- get_kmeans() trains or loads the K-Means model.
- get_cluster_summary() calculates customer count, average income, average spending, and gender distribution for each cluster.
- get_silhouette() prepares validation data for the clustering quality section.
- render_template() sends all values to the front-end clustering page.

CODE SNIPPETS AND EXPLANATION

Elbow Method and Silhouette Logic

```
for k in range(1, 11):

    km = KMeans(n_clusters=k, init="k-means++", n_init=10, random_state=42)

    km.fit(X_scaled)

    wcss.append(km.inertia_)

for k in range(2, 11):

    km = KMeans(n_clusters=k, init="k-means++", n_init=10, random_state=42)

    labels = km.fit_predict(X_scaled)

    score = silhouette_score(X_scaled, labels)

    scores.append(score)
```

The first loop calculates WCSS for different K values. WCSS shows how compact the clusters are. The second loop calculates silhouette score for each K from 2 to 10. The model finally selects K = 5 because it gives useful business segments with a strong validation score.

Database and Prediction Storage

When a new customer is predicted from the dashboard, the application can store the customer details and predicted cluster in SQLite. This allows the dashboard to maintain a history of predictions and show previously entered customers.

WEBSITE SCREENSHOTS

Home Page / Project Landing Page

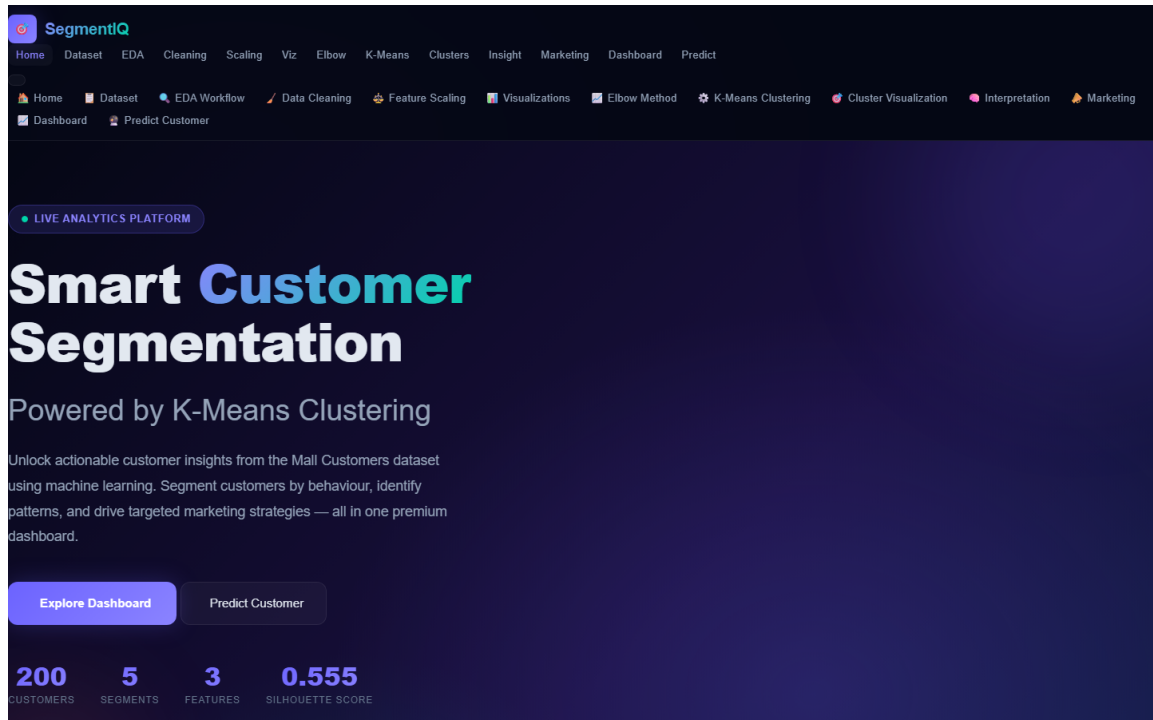


Figure 1: Live Flask dashboard home page showing the customer segmentation platform overview.

WEBSITE SCREENSHOTS

Dataset Overview Page

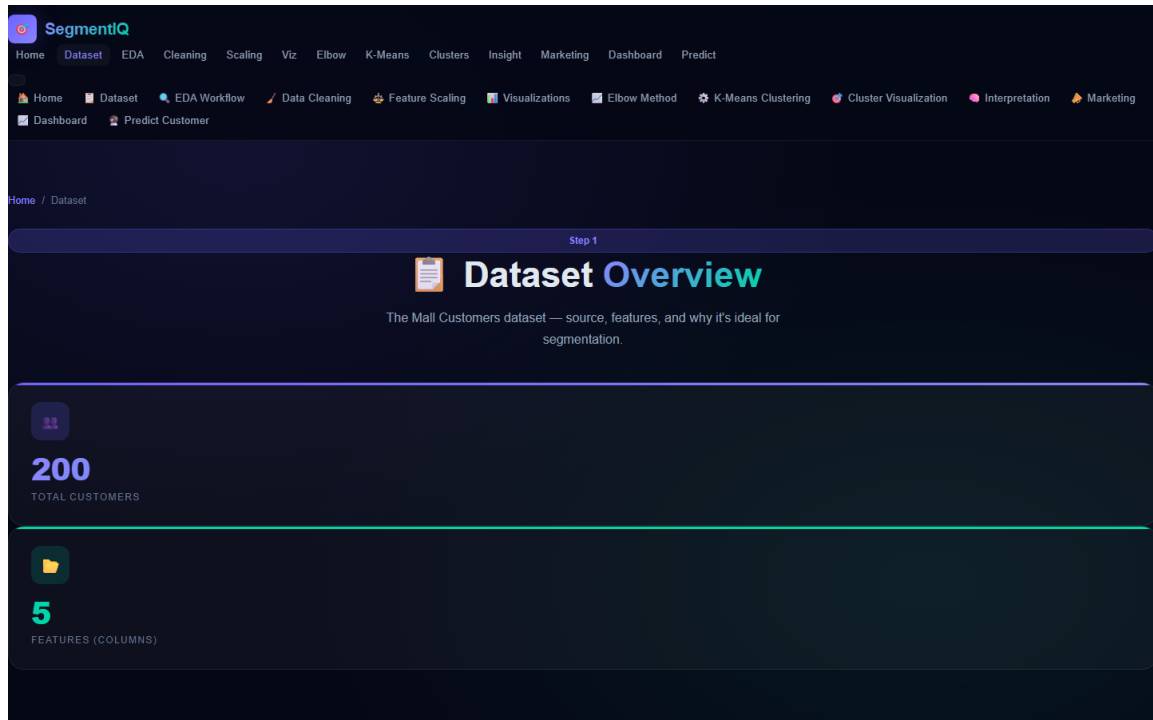


Figure 2: Dataset page showing records, columns, and preview of the Mall Customers dataset.

WEBSITE SCREENSHOTS

EDA Workflow Page

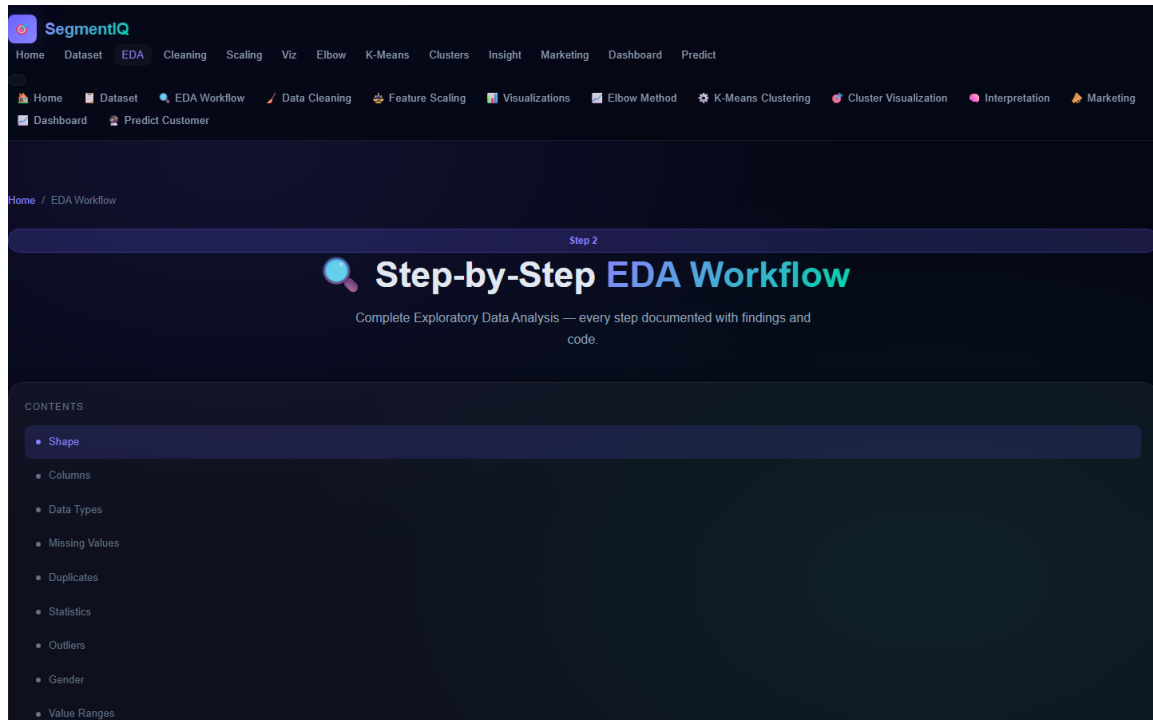


Figure 3: EDA page showing exploratory analysis steps used before clustering.

WEBSITE SCREENSHOTS

Visualizations Page

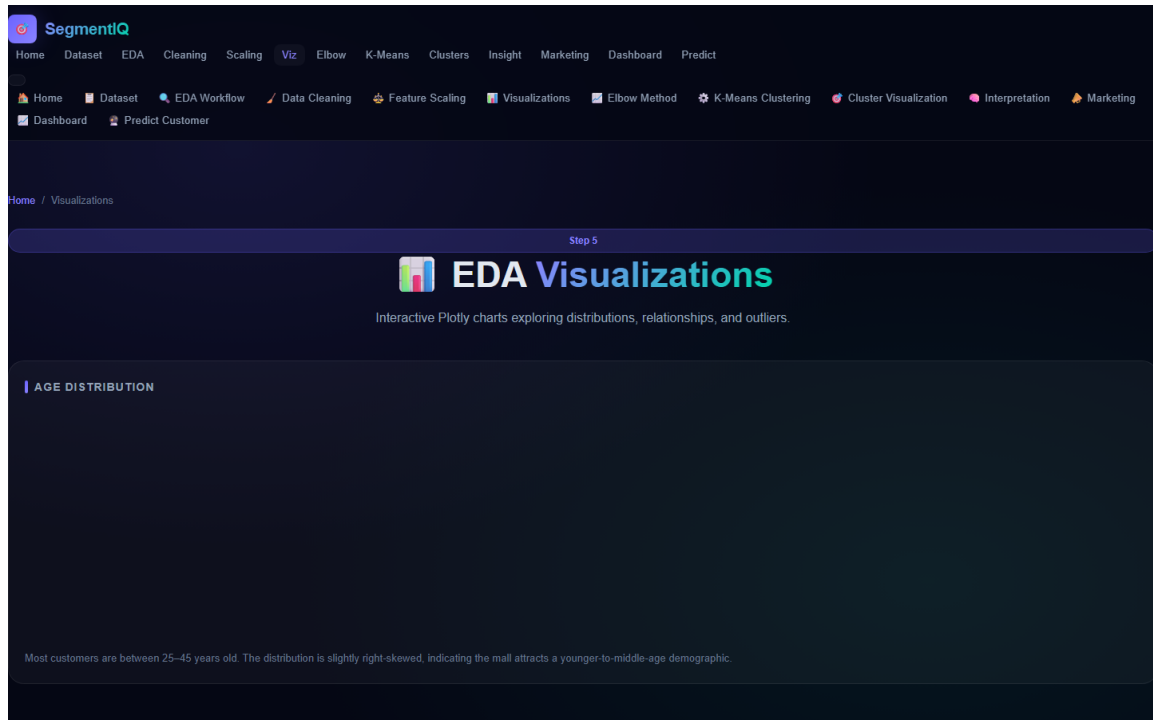


Figure 4: Visualization page containing interactive charts for understanding customer behaviour.

WEBSITE SCREENSHOTS

K-Means Clustering Page

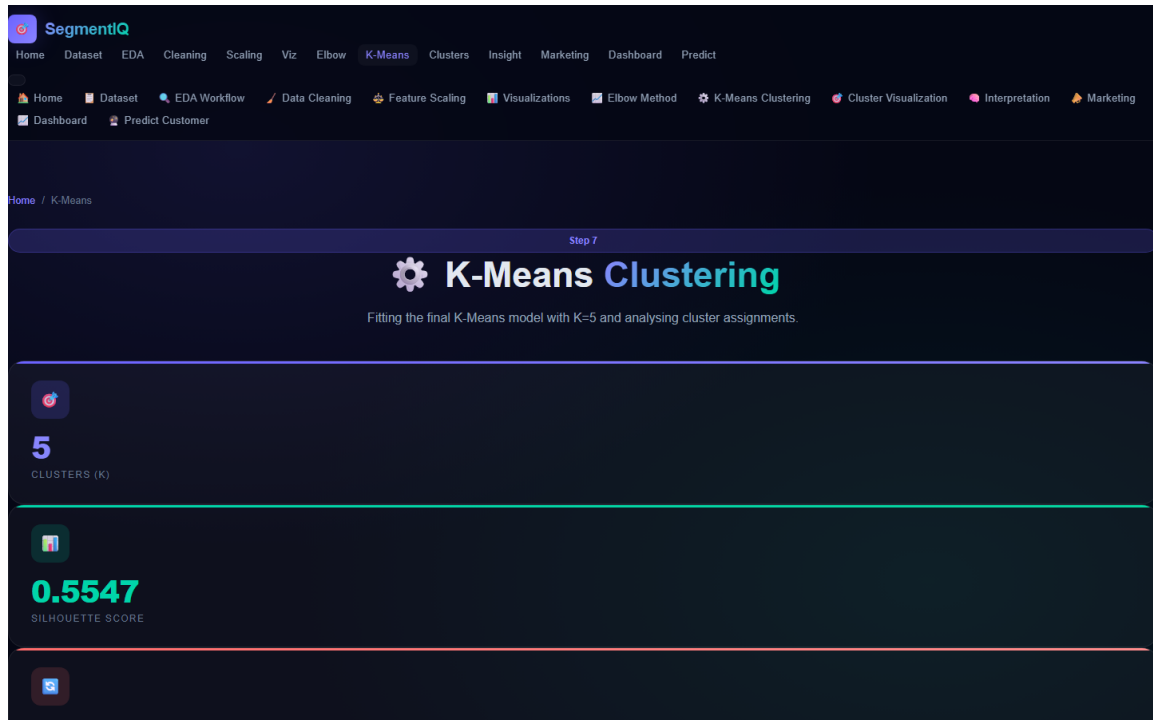


Figure 5: K-Means page showing model configuration, cluster quality, and summary output.

WEBSITE SCREENSHOTS

Cluster Visualization Page

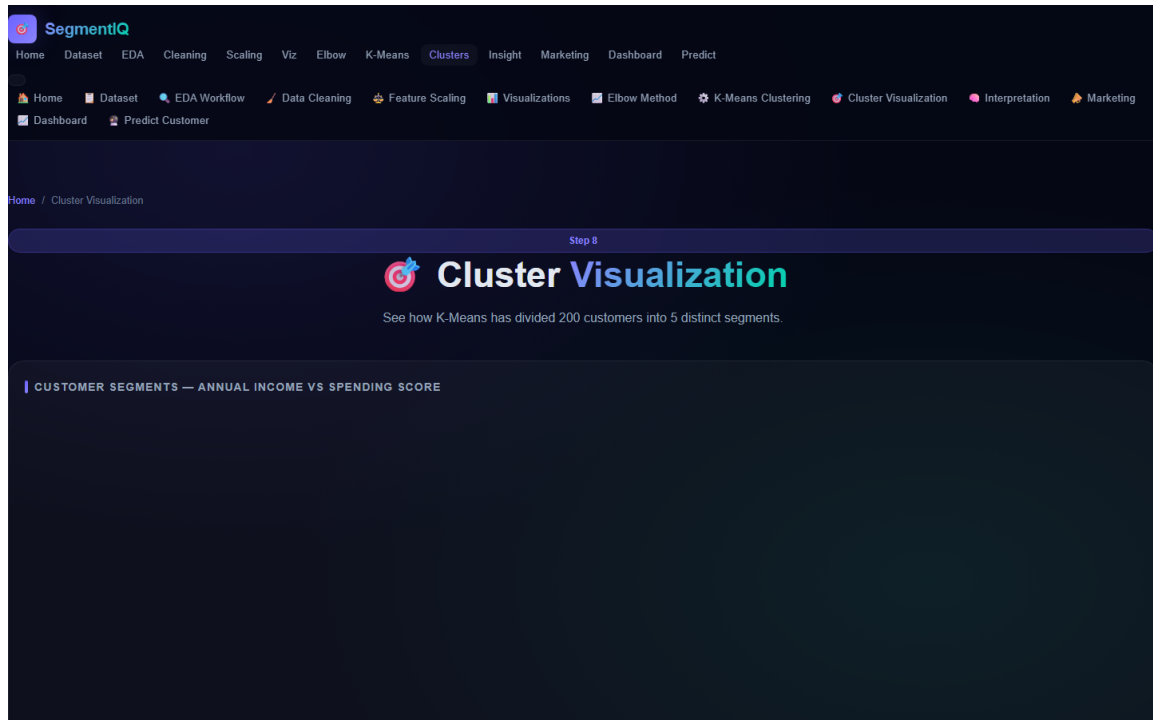


Figure 6: Cluster visualization page showing how customers are divided into segments.

WEBSITE SCREENSHOTS

Marketing Recommendations Page

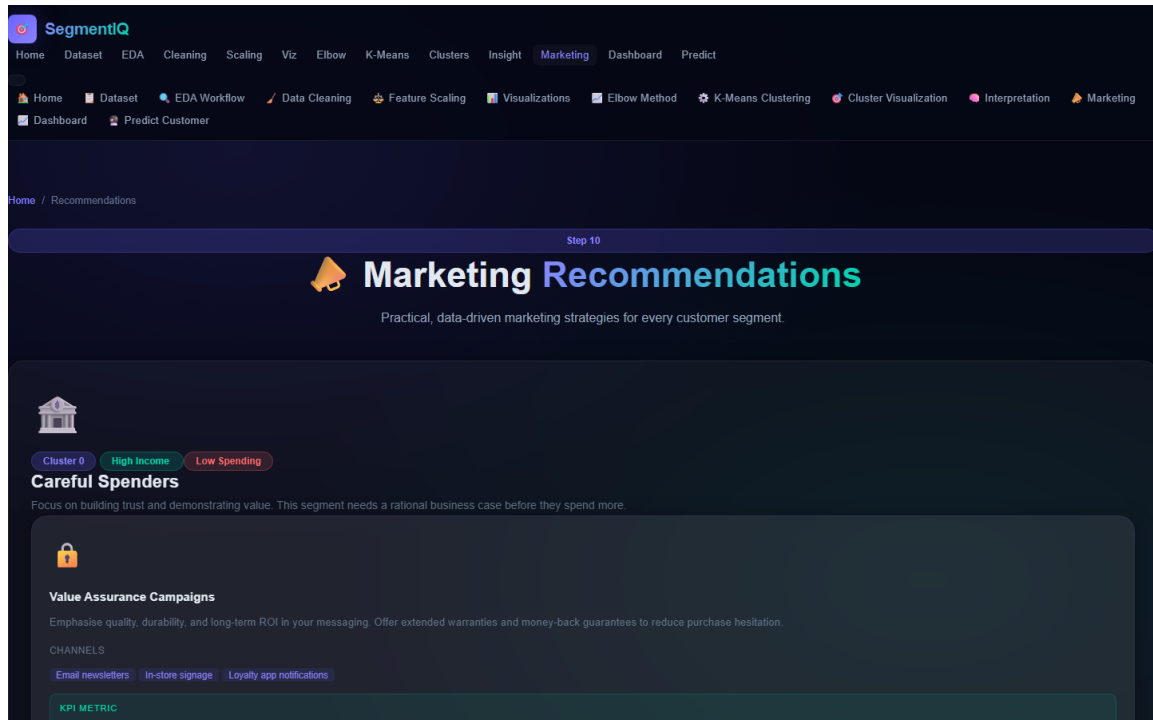


Figure 7: Recommendation page showing business strategies for each segment.

WEBSITE SCREENSHOTS

Dashboard Page

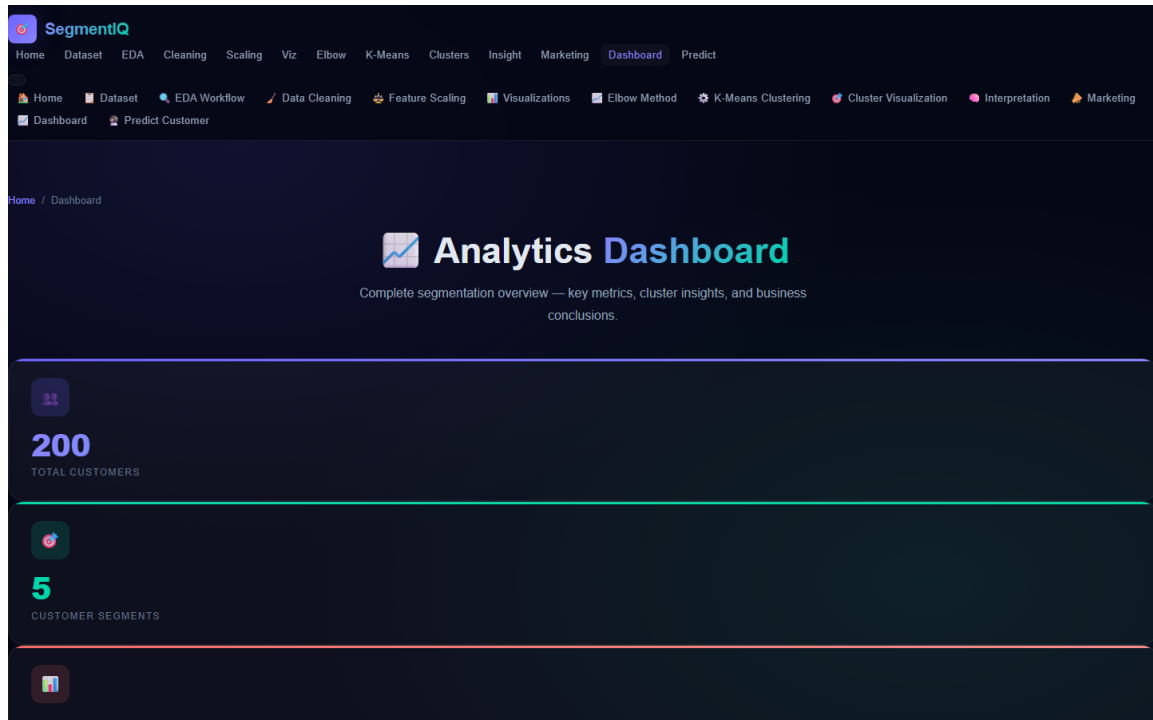


Figure 8: Dashboard page summarizing segmentation results and saved customer predictions.

WEBSITE SCREENSHOTS

Predict Customer Page

SegmentIQ

Home Dataset EDA Cleaning Scaling Viz Elbow K-Means Clusters Insight Marketing Dashboard Predict

Home Dataset EDA Workflow Data Cleaning Feature Scaling Visualizations Elbow Method K-Means Clustering Cluster Visualization Interpretation Marketing

Dashboard Predict Customer

Home / Predict

Predict Customer Segment

Enter customer details to predict their segment using the trained K-Means model.

Enter Customer Details

Customer Name (optional) e.g. John Smith

Gender Male

Age 30

Annual Income (k\$) \$60k

\$10k \$150k

Spending Score (1-100) 50/100

1 (Low) 100 (High)

Predict Segment

How it works:
Your input is scaled using the saved StandardScaler, then fed into the trained K-Means model to assign the nearest cluster centroid. The result is saved to the database.

Figure 9: Prediction page where a new customer's segment can be predicted using income and spending score.

MODEL OUTPUT CHARTS

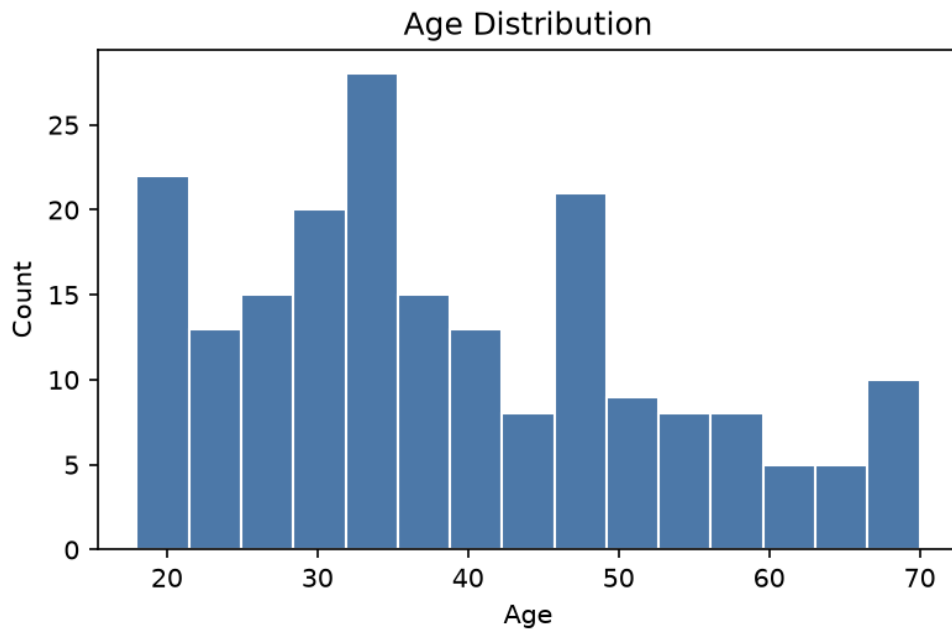


Figure 10: Age distribution of customers



Figure 11: Annual income versus spending score

MODEL OUTPUT CHARTS

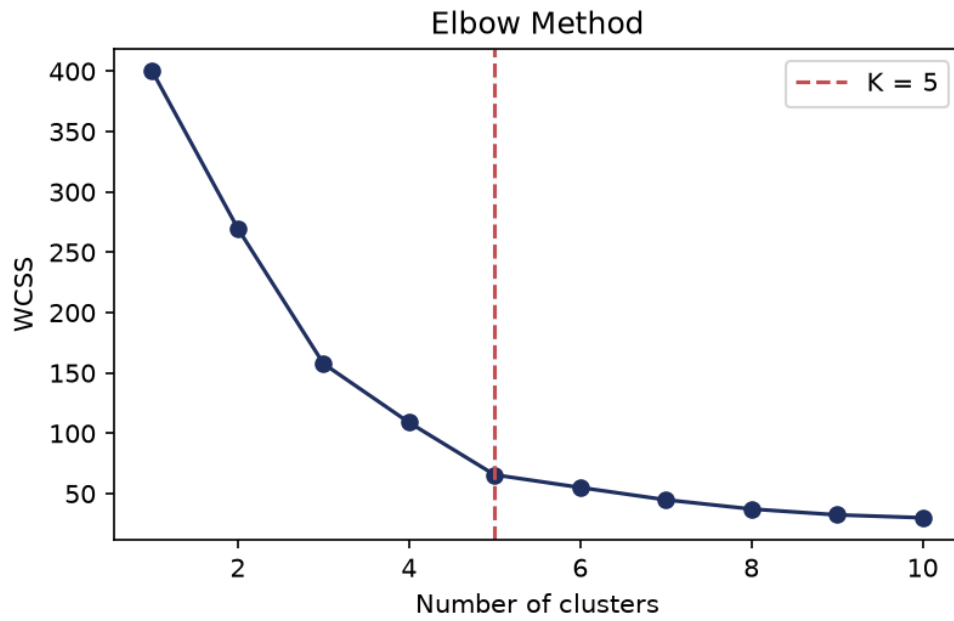


Figure 12: Elbow method for selecting number of clusters

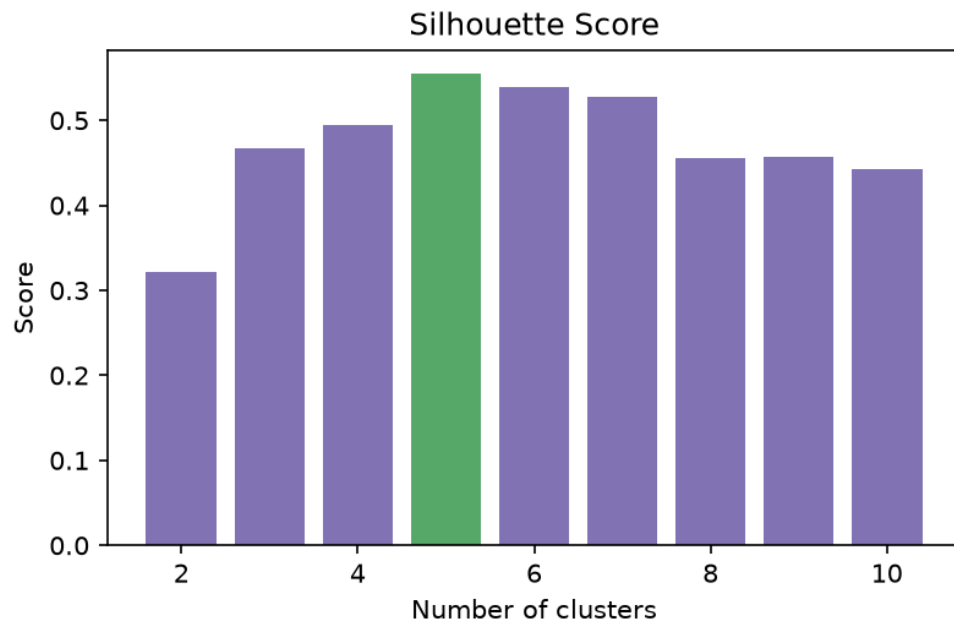


Figure 13: Silhouette score comparison

MODEL OUTPUT CHARTS

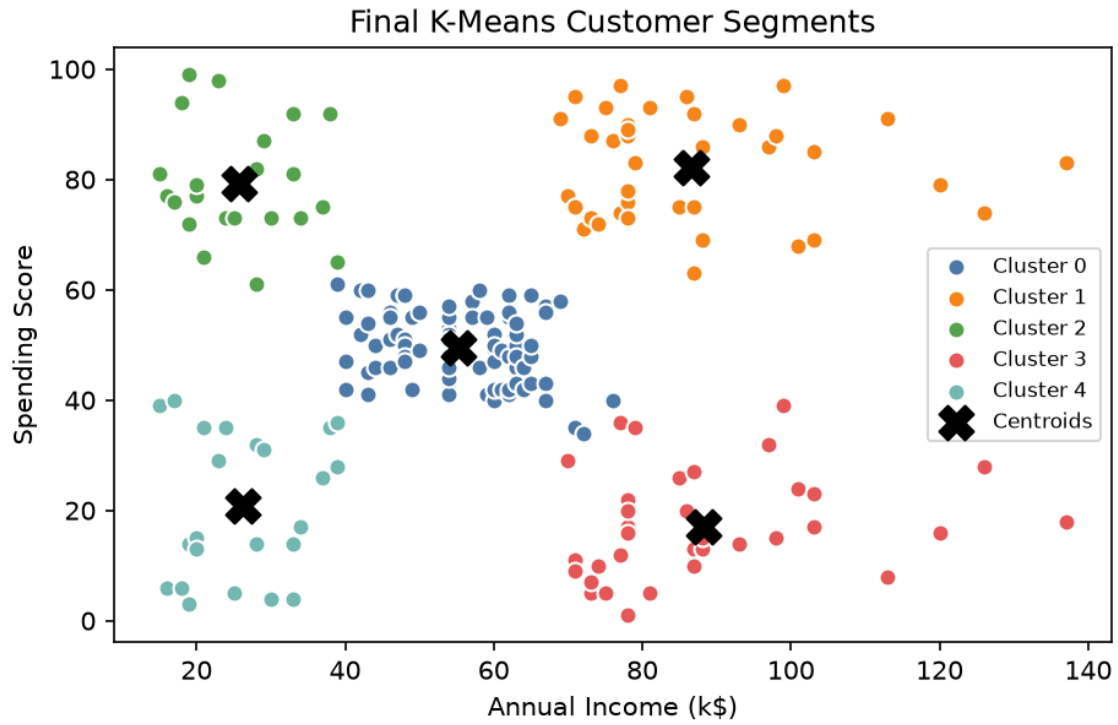


Figure 14: Final K-Means cluster visualization

Cluster	Segment	Size	Avg Income	Avg Spending
0	Average Customers	81	55.3	49.5
1	Target Customers	39	86.5	82.1
2	Impulsive Buyers	22	25.7	79.4
3	Careful Spenders	35	88.2	17.1
4	Budget Shoppers	23	26.3	20.9

MODEL OUTPUT CHARTS

Sample Dataset Preview

CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40
6	Female	22	17	76
7	Female	35	18	6
8	Female	23	18	94

The dashboard pages show dataset overview, EDA, cleaning, scaling, clustering results, interpretation, recommendations, and new customer prediction.

TESTING

Test Case	Expected Result	Status
Dataset loads successfully	CSV contains 200 records	Pass
Missing value check	No missing values after cleaning	Pass
Scaling check	Selected features transformed by StandardScaler	Pass
Elbow method	WCSS decreases as K increases	Pass
Silhouette score	Model returns score above 0.5 for K = 5	Pass
Prediction function	New customer receives a cluster label	Pass
Dashboard routes	Pages render project outputs	Pass

RESULTS

The K-Means model successfully segmented all 200 customers into five groups. The final silhouette score is 0.5547, which indicates good cluster separation for this dataset.

Cluster	Segment Name	Customers	Share	Avg Age	Male / Female
0	Average Customers	81	40.5%	42.7	33 / 48
1	Target Customers	39	19.5%	32.7	18 / 21
2	Impulsive Buyers	22	11.0%	25.3	9 / 13
3	Careful Spenders	35	17.5%	41.1	19 / 16
4	Budget Shoppers	23	11.5%	45.2	9 / 14

The segments provide direct business value. High-income high-spending customers can be treated as premium target customers, while high-income low-spending customers can be encouraged through value-based offers. Low-income high-spending customers respond well to trend-based campaigns, and low-income low-spending customers benefit from discount and loyalty programs.

CONCLUSION

The Customer Segmentation Assistant successfully demonstrates a complete AI-ML workflow for a real business analytics use case. The project starts from raw customer data, performs EDA and preprocessing, applies K-Means clustering, and presents the output through visual dashboards and a formal report.

The final model identifies five meaningful customer groups. These groups can help a retail business plan personalized promotions, improve customer engagement, and make data-driven marketing decisions.

FUTURE WORK

- Use larger real-world customer transaction datasets.
- Add product categories, visit frequency, and purchase history as features.
- Compare K-Means with hierarchical clustering, DBSCAN, and Gaussian mixture models.
- Deploy the Flask dashboard on a cloud platform.
- Integrate real-time CRM data for continuously updated segmentation.

REFERENCES

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